

CSE 3203 – Software Engineering

Lecture 09: Requirements Engineering – Contd.

Atanu Shuvam Roy



Lecturer,
Dept. of CSE, ULAB, Dhaka



EMAIL:
atanu.shuvam@ulab.edu.bd



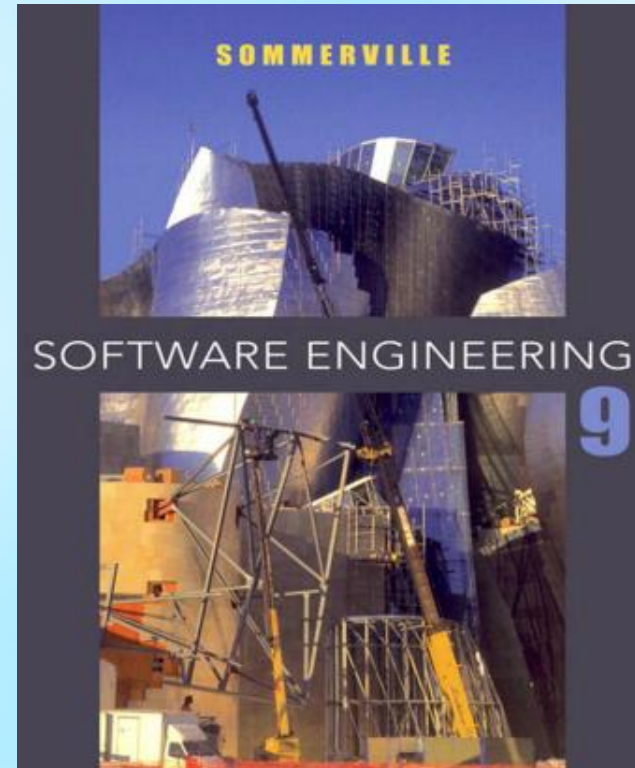
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Book for Reference

- Software Engineering 9th Edition by Ian Sommerville



Mark Distribution



- Mid Semester Exam – 25%
- Final Semester Exam – 40%
- Class Attendance & Performance – 10%
- Course Project – 25%

Is it possible to get A/A+ ?

- Absolutely. Just follow my instructions to the point.

How to fail in this course ?

- Don't follow my instructions.

Instructions



Topics covered

- ✧ Functional and non-functional requirements
- ✧ The software requirements document
- ✧ Requirements specification
- ✧ Requirements engineering processes
- ✧ Requirements elicitation and analysis
- ✧ Requirements validation
- ✧ Requirements management

Structured specifications

- ✧ An approach to writing requirements where the freedom of the requirements writer is limited and requirements are written in a standard way.
- ✧ This works well for some types of requirements e.g. requirements for embedded control system but is sometimes too rigid for writing business system requirements.

Form-based specifications

- ✧ Definition of the function or entity.
- ✧ Description of inputs and where they come from.
- ✧ Description of outputs and where they go to.
- ✧ Information about the information needed for the computation and other entities used.
- ✧ Description of the action to be taken.
- ✧ Pre and post conditions (if appropriate).
- ✧ The side effects (if any) of the function.

A structured specification of a requirement for an insulin pump

Insulin Pump/Control Software/SRS/3.3.2

Function Compute insulin dose: safe sugar level.

Description

Computes the dose of insulin to be delivered when the current measured sugar level is in the safe zone between 3 and 7 units.

Inputs Current sugar reading (r2); the previous two readings (r0 and r1).

Source Current sugar reading from sensor. Other readings from memory.

Outputs CompDose—the dose in insulin to be delivered.

Destination Main control loop.

A structured specification of a requirement for an insulin pump

Action

CompDose is zero if the sugar level is stable or falling or if the level is increasing but the rate of increase is decreasing. If the level is increasing and the rate of increase is increasing, then CompDose is computed by dividing the difference between the current sugar level and the previous level by 4 and rounding the result. If the result, is rounded to zero then CompDose is set to the minimum dose that can be delivered.

Requirements

Two previous readings so that the rate of change of sugar level can be computed.

Pre-condition

The insulin reservoir contains at least the maximum allowed single dose of insulin.

Post-condition r_0 is replaced by r_1 then r_1 is replaced by r_2 .

Side effects None.

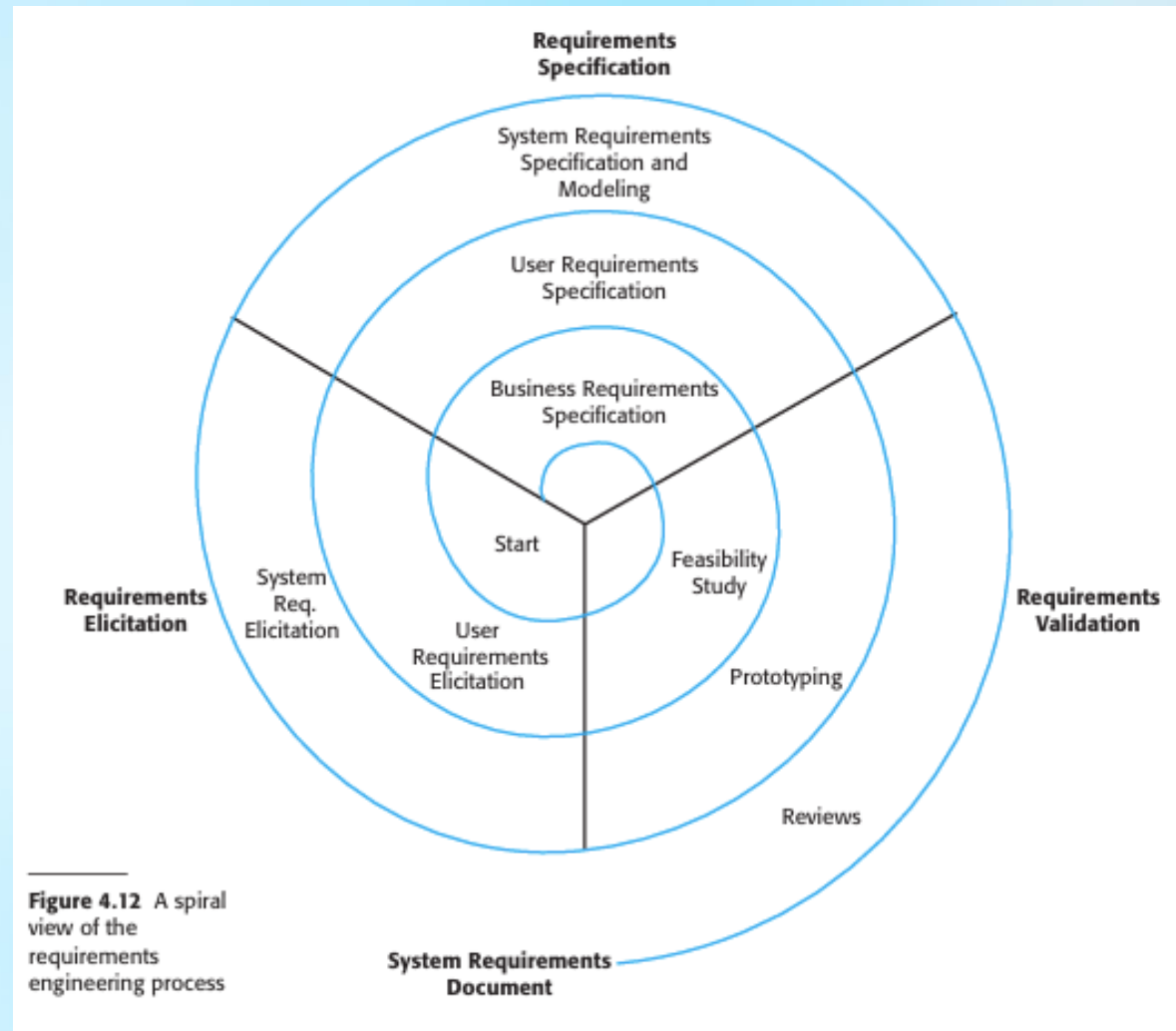
Tabular specification

- ✧ Used to supplement natural language.
- ✧ Particularly useful when you have to define a number of possible alternative courses of action.
- ✧ For example, the insulin pump systems bases its computations on the rate of change of blood sugar level and the tabular specification explains how to calculate the insulin requirement for different scenarios.

Tabular specification of computation for an insulin pump

Condition	Action
Sugar level falling ($r2 < r1$)	CompDose = 0
Sugar level stable ($r2 = r1$)	CompDose = 0
Sugar level increasing and rate of increase decreasing ($(r2 - r1) < (r1 - r0)$)	CompDose = 0
Sugar level increasing and rate of increase stable or increasing ($(r2 - r1) \geq (r1 - r0)$)	CompDose = round $((r2 - r1)/4)$ If rounded result = 0 then CompDose = MinimumDose

A spiral view of the requirements engineering process



Requirements engineering processes

- ✧ The processes used for RE vary widely depending on the application domain, the people involved and the organisation developing the requirements.
- ✧ However, there are a number of generic activities common to all processes
 - Requirements elicitation;
 - Requirements analysis;
 - Requirements validation;
 - Requirements management.
- ✧ In practice, RE is an iterative activity in which these processes are interleaved.

Requirements elicitation and analysis

- ✧ Sometimes called requirements elicitation or requirements discovery.
- ✧ Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints.
- ✧ May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called *stakeholders*.

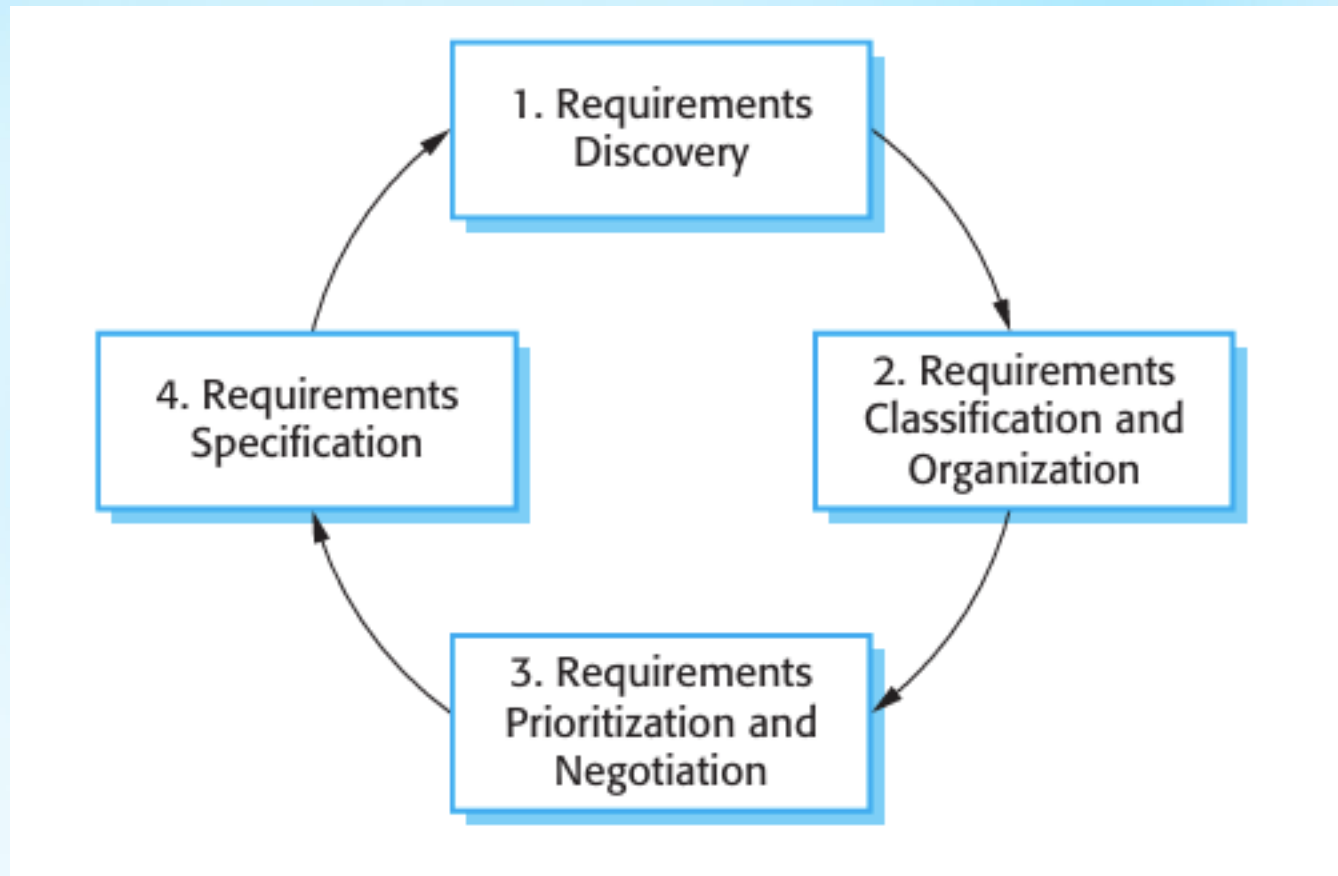
Problems of requirements analysis

- ✧ Stakeholders don't know what they really want.
- ✧ Stakeholders express requirements in their own terms.
- ✧ Different stakeholders may have conflicting requirements.
- ✧ Organisational and political factors may influence the system requirements.
- ✧ The requirements change during the analysis process. New stakeholders may emerge and the business environment may change.

Requirements elicitation and analysis

- ✧ Software engineers work with a range of system stakeholders to find out about the application domain, the services that the system should provide, the required system performance, hardware constraints, other systems, etc.
- ✧ Stages include:
 - Requirements discovery,
 - Requirements classification and organization,
 - Requirements prioritization and negotiation,
 - Requirements specification.

The requirements elicitation and analysis process



Process activities

✧ Requirements discovery

- Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.

✧ Requirements classification and organisation

- Groups related requirements and organises them into coherent clusters.

✧ Prioritisation and negotiation

- Prioritising requirements and resolving requirements conflicts.

✧ Requirements specification

- Requirements are documented and input into the next round of the spiral.

Problems of requirements elicitation

- ✧ Stakeholders don't know what they really want.
- ✧ Stakeholders express requirements in their own terms.
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- ✧ The requirements change during the analysis process. New stakeholders may emerge and the business environment change.

Requirements discovery

- ✧ The process of gathering information about the required and existing systems and distilling the user and system requirements from this information.
- ✧ Interaction is with system stakeholders from managers to external regulators.
- ✧ Systems normally have a range of stakeholders.

Stakeholders in the MHC-PMS

- ✧ Patients whose information is recorded in the system.
- ✧ Doctors who are responsible for assessing and treating patients.
- ✧ Nurses who coordinate the consultations with doctors and administer some treatments.
- ✧ Medical receptionists who manage patients' appointments.
- ✧ IT staff who are responsible for installing and maintaining the system.

Stakeholders in the MHC-PMS

- ✧ A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
- ✧ Health care managers who obtain management information from the system.
- ✧ Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

Interviewing

✧ Formal or informal interviews with stakeholders are part of most RE processes.

✧ Types of interview

- Closed interviews based on pre-determined list of questions
- Open interviews where various issues are explored with stakeholders.

✧ Effective interviewing

- Be open-minded, avoid pre-conceived ideas about the requirements and are willing to listen to stakeholders.
- Prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

Interviews in practice

- ✧ Normally a mix of closed and open-ended interviewing.
- ✧ Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- ✧ Interviews are not good for understanding domain requirements
 - Requirements engineers cannot understand specific domain terminology;
 - Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

Scenarios

- ✧ Scenarios are real-life examples of how a system can be
 - used.
- ✧ They should include
 - A description of the starting situation;
 - A description of the normal flow of events;
 - A description of what can go wrong;
 - Information about other concurrent activities;
 - A description of the state when the scenario finishes.

Scenario for collecting medical history in MHC-PMS

Initial assumption: The patient has seen a medical receptionist who has created a record in the system and collected the patient's personal information (name, address, age, etc.). A nurse is logged on to the system and is collecting medical history.

Normal: The nurse searches for the patient by family name. If there is more than one patient with the same surname, the given name (first name in English) and date of birth are used to identify the patient.

The nurse chooses the menu option to add medical history.

The nurse then follows a series of prompts from the system to enter information about consultations elsewhere on mental health problems (free text input), existing medical conditions (nurse selects conditions from menu), medication currently taken (selected from menu), allergies (free text), and home life (form).

Scenario for collecting medical history in MHC-PMS

What can go wrong: The patient's record does not exist or cannot be found. The nurse should create a new record and record personal information.

Patient conditions or medication are not entered in the menu. The nurse should choose the 'other' option and enter free text describing the condition/medication.

Patient cannot/will not provide information on medical history. The nurse should enter free text recording the patient's inability/unwillingness to provide information. The system should print the standard exclusion form stating that the lack of information may mean that treatment will be limited or delayed. This should be signed and handed to the patient.

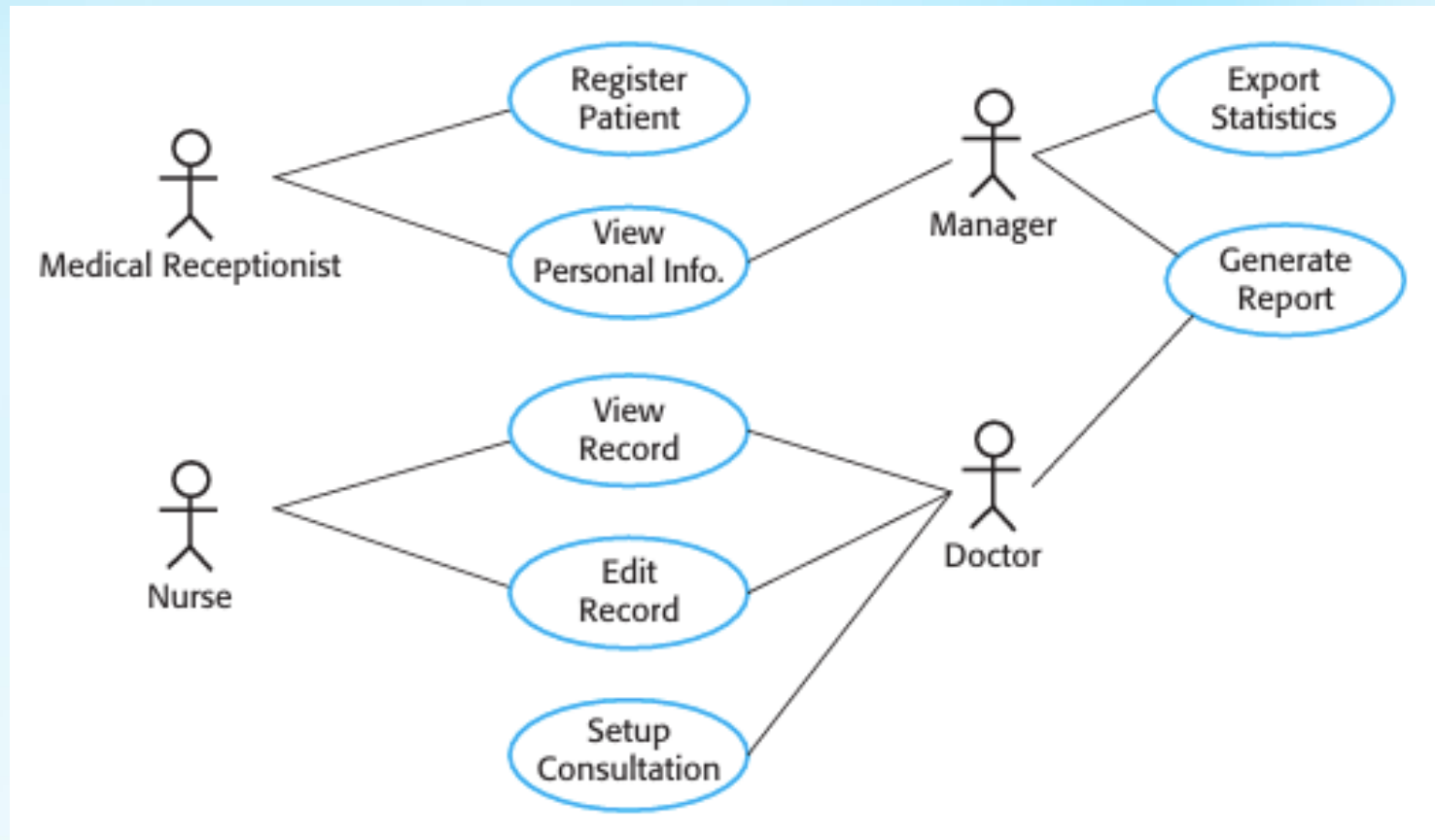
Other activities: Record may be consulted but not edited by other staff while information is being entered.

System state on completion: User is logged on. The patient record including medical history is entered in the database, a record is added to the system log showing the start and end time of the session and the nurse involved.

Use cases

- ✧ Use-cases are a scenario based technique in the UML which identify the actors in an interaction and which describe the interaction itself.
- ✧ A set of use cases should describe all possible interactions with the system.
- ✧ High-level graphical model supplemented by more detailed tabular description (see Chapter 5).
- ✧ Sequence diagrams may be used to add detail to use-cases by showing the sequence of event processing in the system.

Use cases for the MHC-PMS



Ethnography

- ✧ A social scientist spends a considerable time observing and analysing how people actually work.
- ✧ People do not have to explain or articulate their work.
- ✧ Social and organisational factors of importance may be observed.
- ✧ Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models.

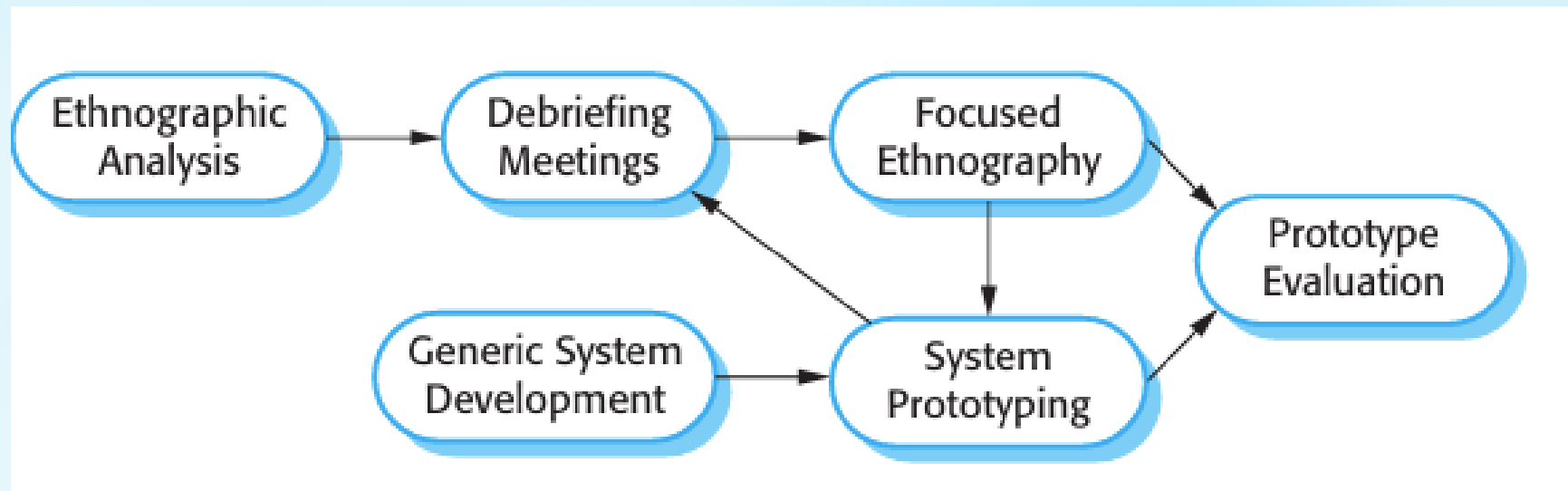
Scope of ethnography

- ✧ Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work.
- ✧ Requirements that are derived from cooperation and awareness of other people's activities.
 - Awareness of what other people are doing leads to changes in the ways in which we do things.
- ✧ Ethnography is effective for understanding existing processes but cannot identify new features that should be added to a system.

Focused ethnography

- ✧ Developed in a project studying the air traffic control process
- ✧ Combines ethnography with prototyping
- ✧ Prototype development results in unanswered questions which focus the ethnographic analysis.
- ✧ The problem with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant.

Ethnography and prototyping for requirements analysis



Requirements validation

- ✧ Concerned with demonstrating that the requirements define the system that the customer really wants.
- ✧ Requirements error costs are high so validation is very important
 - Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.

Requirements checking

- ✧ **Validity**. Does the system provide the functions which best support the customer's needs?
- ✧ **Consistency**. Are there any requirements conflicts?
- ✧ **Completeness**. Are all functions required by the customer included?
- ✧ **Realism**. Can the requirements be implemented given available budget and technology
- ✧ **Verifiability**. Can the requirements be checked?

Requirements validation techniques

✧ Requirements reviews

- Systematic manual analysis of the requirements.

✧ Prototyping

- Using an executable model of the system to check requirements.
Covered in Chapter 2.

✧ Test-case generation

- Developing tests for requirements to check testability.

Requirements reviews

- ✧ Regular reviews should be held while the requirements definition is being formulated.
- ✧ Both client and contractor staff should be involved in reviews.
- ✧ Reviews may be formal (with completed documents) or informal. Good communications between developers, customers and users can resolve problems at an early stage.

Review checks

✧ Verifiability

- Is the requirement realistically testable?

✧ Comprehensibility

- Is the requirement properly understood?

✧ Traceability

- Is the origin of the requirement clearly stated?

✧ Adaptability

- Can the requirement be changed without a large impact on other requirements?

Requirements management

- ✧ Requirements management is the process of managing changing requirements during the requirements engineering process and system development.
- ✧ New requirements emerge as a system is being developed and after it has gone into use.
- ✧ You need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes. You need to establish a formal process for making change proposals and linking these to system requirements.

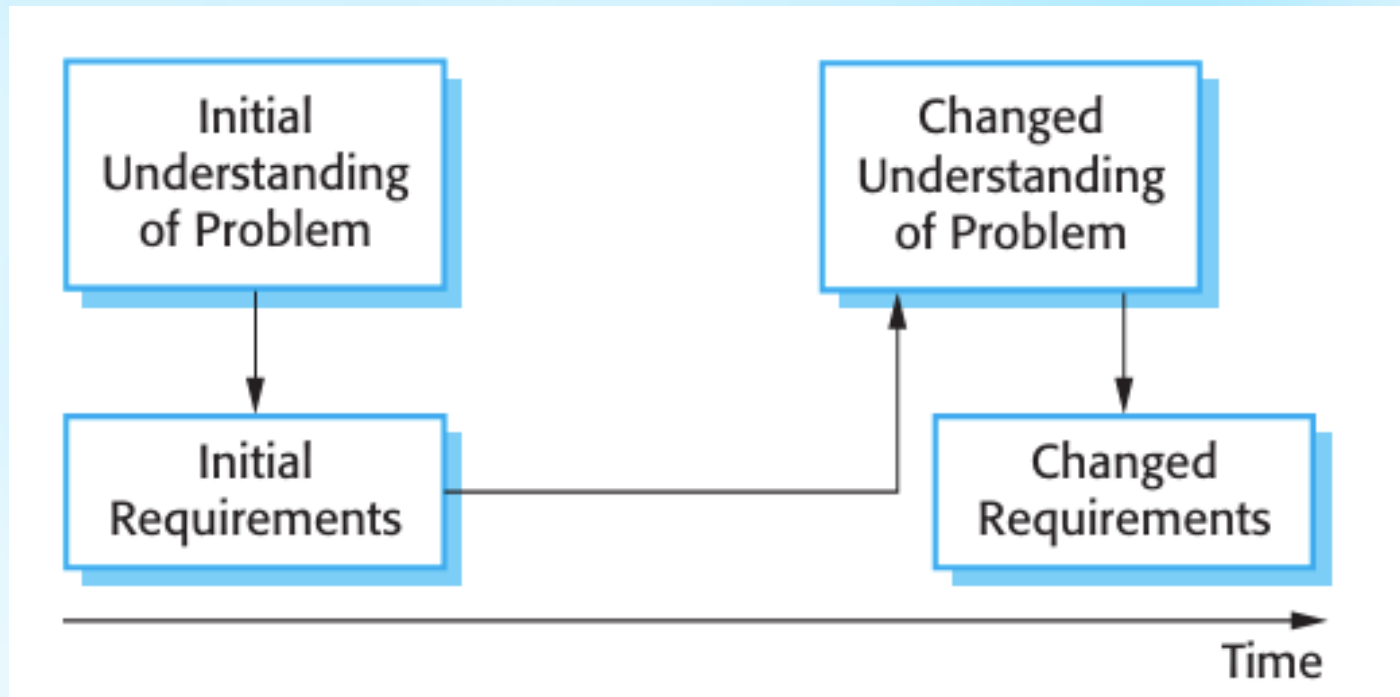
Changing requirements

- ✧ The business and technical environment of the system always changes after installation.
 - New hardware may be introduced, it may be necessary to interface the system with other systems, business priorities may change (with consequent changes in the system support required), and new legislation and regulations may be introduced that the system must necessarily abide by.
- ✧ The people who pay for a system and the users of that system are rarely the same people.
 - System customers impose requirements because of organizational and budgetary constraints. These may conflict with end-user requirements and, after delivery, new features may have to be added for user support if the system is to meet its goals.

Changing requirements

- ✧ Large systems usually have a diverse user community, with many users having different requirements and priorities that may be conflicting or contradictory.
 - The final system requirements are inevitably a compromise between them and, with experience, it is often discovered that the balance of support given to different users has to be changed.

Requirements evolution



Requirements management planning

- ✧ Establishes the level of requirements management detail that is required.
- ✧ Requirements management decisions:
 - *Requirements identification* Each requirement must be uniquely identified so that it can be cross-referenced with other requirements.
 - *A change management process* This is the set of activities that assess the impact and cost of changes. I discuss this process in more detail in the following section.
 - *Traceability policies* These policies define the relationships between each requirement and between the requirements and the system design that should be recorded.
 - *Tool support* Tools that may be used range from specialist requirements management systems to spreadsheets and simple database systems.

Requirements change management

✧ Deciding if a requirements change should be accepted

▪ *Problem analysis and change specification*

- During this stage, the problem or the change proposal is analyzed to check that it is valid. This analysis is fed back to the change requestor who may respond with a more specific requirements change proposal, or decide to withdraw the request.

▪ *Change analysis and costing*

- The effect of the proposed change is assessed using traceability information and general knowledge of the system requirements. Once this analysis is completed, a decision is made whether or not to proceed with the requirements change.

▪ *Change implementation*

- The requirements document and, where necessary, the system design and implementation, are modified. Ideally, the document should be organized so that changes can be easily implemented.

Requirements change management



Key points

- ✧ Requirements for a software system set out what the system should do and define constraints on its operation and implementation.
- ✧ Functional requirements are statements of the services that the system must provide or are descriptions of how some computations must be carried out.
- ✧ Non-functional requirements often constrain the system being developed and the development process being used.
- ✧ They often relate to the emergent properties of the system and therefore apply to the system as a whole.

Key points

- ✧ The software requirements document is an agreed statement of the system requirements. It should be organized so that both system customers and software developers can use it.
- ✧ The requirements engineering process is an iterative process including requirements elicitation, specification and validation.
- ✧ Requirements elicitation and analysis is an iterative process that can be represented as a spiral of activities – requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation.

Key points

- ✧ You can use a range of techniques for requirements elicitation including interviews, scenarios, use-cases and ethnography.
- ✧ Requirements validation is the process of checking the requirements for validity, consistency, completeness, realism and verifiability.
- ✧ Business, organizational and technical changes inevitably lead to changes to the requirements for a software system. Requirements management is the process of managing and controlling these changes.



Thank You

atanu.Shuvam@ulab.edu.bd

University of Liberal Arts Bangladesh
Department of Computer Science & Engineering